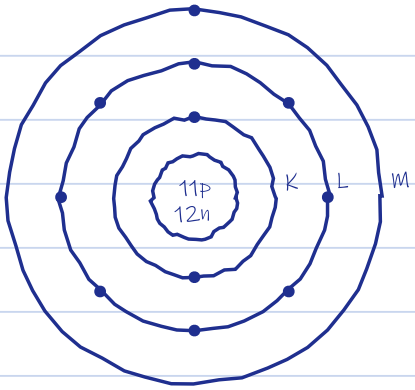
Name: Ishaan Mehta Roll No: 1CH24CS011 Sec: B

Chemistry - Internal Assessment 1

Ans 1)

Na atom: Z = 11, A = 23, so the nucleus has 11 protons and 23 - 11 = 12 neutrons.

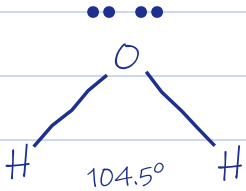


Sodium (Na)

Electronic configuration = 2, 8, 1 (1s2 2s2 2p6 3s1). Valency = 1 as Na loses its one outer electron to form Na+.

Ans 2)

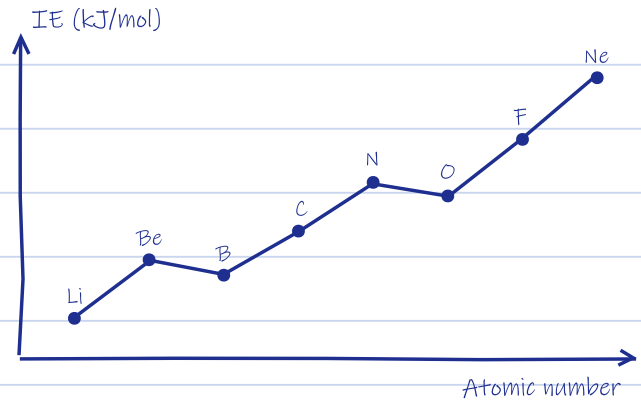
Lewis structure: O has 2 bond pairs with the two H atoms and 2 lone pairs.



bent / V shape

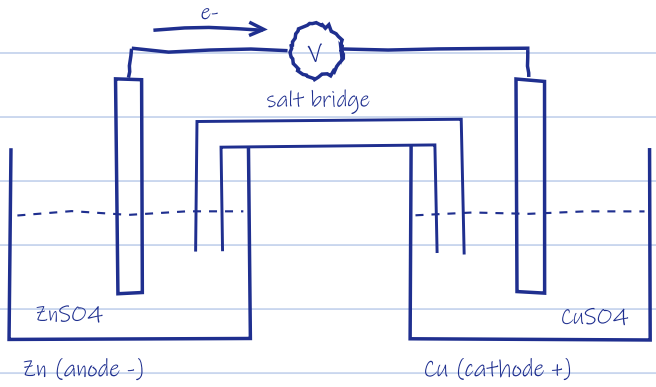
Bond angle is 104.5°. The lone pair - lone pair repulsion is more than bond pair - bond pair repulsion, so the O-H bonds are pushed closer and the angle becomes less than 109.5°.

Ans 3)



IE increases from Li to Ne because nuclear charge increases and atomic size decreases, so electrons are held more tightly. B is lower than Be because the electron is removed from 2p which is higher in energy and shielded by 2s. O is lower than N because in O the 2p electron is paired and the repulsion makes it easier to remove.

Ans 4)

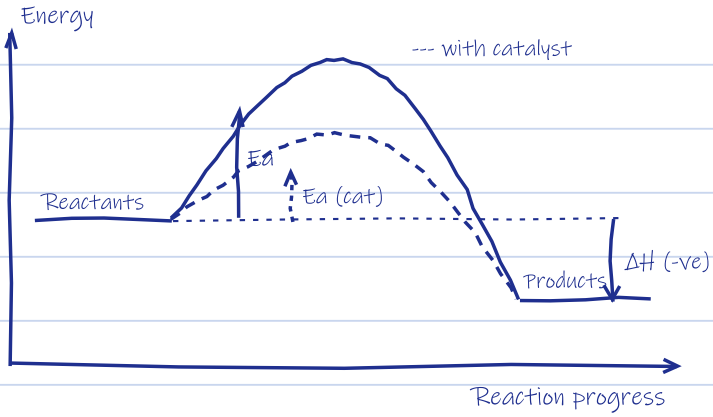


Anode: Zn -> Zn2+ + 2e- (oxidation)

Cathode: Cu2+ + 2e- -> Cu (reduction)

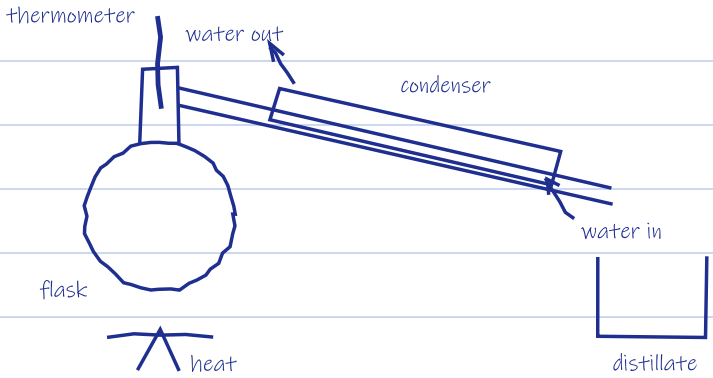
E cell = E cathode - E anode = 0.34 - (-0.76) = 1.10 V

Ans 5)



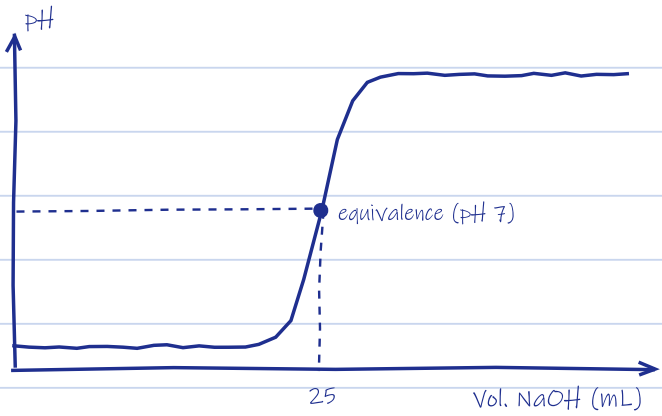
Products are lower so dH is negative (exothermic). A catalyst gives an alternative path with lower activation energy, so more molecules have energy more than Ea and more collisions are successful, so rate increases. dH does not change.

Ans 6)



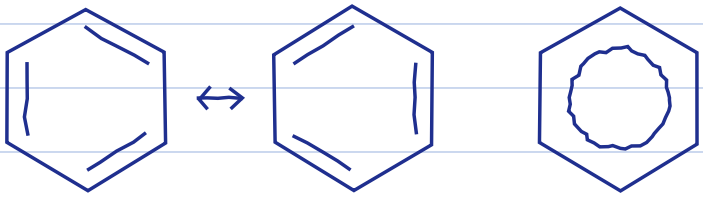
The condenser cools the vapour so it turns back into liquid which is collected as distillate. Water enters at the bottom so that the condenser is completely filled with water and cooling is efficient (counter current flow).

Ans 7)



moles of HCl = 0.10 x 25/1000 = 0.0025 mol. V of NaOH = 0.0025/0.10 = 0.025 L = 25.0 mL. Equivalence point at pH 7. Indicator: phenolphthalein.

Ans 8)

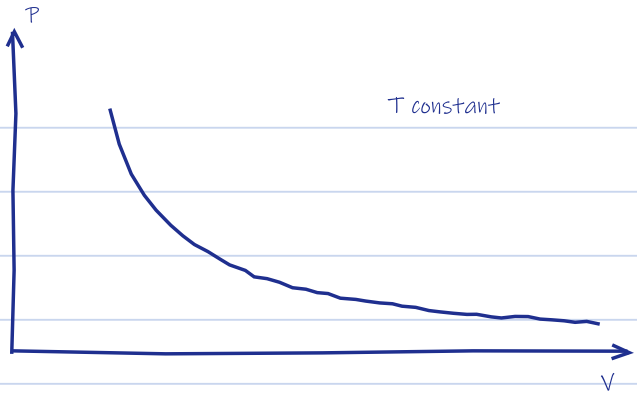


Kekule structures hybrid

In benzene the 6 pi electrons are delocalised over the ring which gives extra stability (resonance energy). Addition would destroy the delocalised aromatic ring but substitution keeps it, so benzene prefers substitution.

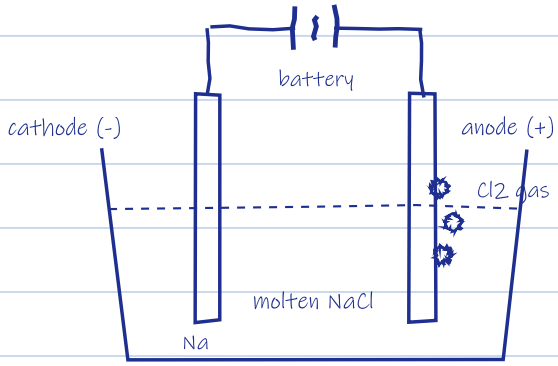
Ans 9)

Boyle's law: at constant temperature the pressure of a fixed mass of gas is inversely proportional to its volume, PV = constant.



P1V1 = P2V2 => 2 x 3 = 6 x V2 => V2 = 1.0 L

Ans 10)



Cathode: Na+ + e- -> Na (sodium metal)

Anode: 2Cl- -> Cl2 + 2e- (chlorine gas)

Sodium is formed at the cathode and chlorine gas at the anode.